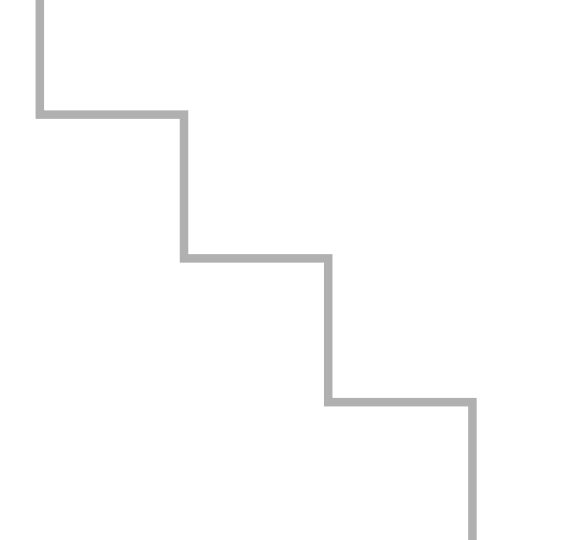


ENERGY-AWARE FEDERATED LEARNING FOR EDGE DEVICES IN LOW- CONNECTIVITY REGIONS



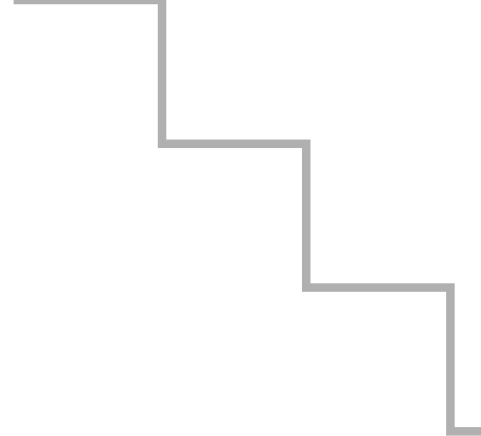
Presented By : Allwyn Noble (URK23CS5050)

Guide : Prof.Neeraja

Department : Department of Artificial Intelligence and Machine Learning

Academic Year : 2026–2027





INTRODUCTION

With the rapid growth of Artificial Intelligence and IoT, more applications are being deployed on edge devices such as smartphones, wearable devices, and sensors. During my research, I found that although Federated Learning provides a privacy-preserving way to train machine learning models, existing approaches perform poorly in real-world scenarios where devices have limited battery life, unstable internet connectivity, and varying computational capabilities.

These challenges inspired me to work on an Energy-Aware Federated Learning framework that uses Deep Reinforcement Learning (Deep Q-Network) to intelligently select the most suitable clients for training. My goal is to improve energy efficiency, reduce communication costs, and enhance model performance in low-connectivity edge environments.



Applications

- Smart Agriculture
- Rural Healthcare
- Disaster Monitoring
- Smart Transportation
- Industrial IoT

Need for Energy-Aware Federated Learning

- Edge devices have limited battery power and computational resources.
 - Network connectivity in rural and remote areas is often unstable.
 - Conventional FL algorithms assume reliable communication, which is impractical in real-world deployments.
- 

PROBLEM STATEMENT

Develop an Energy-Aware Federated Learning framework for edge devices operating in low-connectivity environments.

Build an Android application that performs local model training and collects device telemetry (battery, bandwidth, and network status).

Design a FastAPI server to coordinate federated learning rounds and manage client communications.

Implement a Deep Q-Network (DQN) to intelligently select the most suitable clients based on device conditions.

Integrate FedAvg/FedAsync for aggregating local model updates into a global model.

Evaluate the proposed system using metrics such as:

Model Accuracy

Energy Consumption

Communication Overhead

Convergence Speed

Packet Loss Resilience

RESEARCH GAP

- Existing FL methods assume stable networks.
- Battery, bandwidth, and packet loss are rarely considered together.
- Random client selection leads to energy waste and client dropouts.
- Limited use of DRL for adaptive client selection.
- Need for an energy-aware and intelligent Federated Learning framework.

ALGORITHM USED

Primary Algorithm: Deep Q-Network (DQN)

Purpose:

- Intelligent client selection for Federated Learning
- Chooses the most suitable edge devices for each training round

Input Parameters:

- Battery Level
- Bandwidth
- Packet Loss
- Device Reliability

Decision:

- Selects clients with the best overall conditions

Benefits:

- Reduces energy consumption
- Minimizes communication overhead
- Improves model convergence
- Increases model accuracy

HOW MY PROJECT ADDRESSES THIS GAP

- Uses Deep Q-Network (DQN) to dynamically select the best clients.
- Considers battery level, bandwidth, packet loss, and reliability before each training round.
- Reduces unnecessary communication and energy consumption.
- Improves model accuracy, convergence speed, and training reliability in low-connectivity environments.



OBJECTIVES

Primary Objective

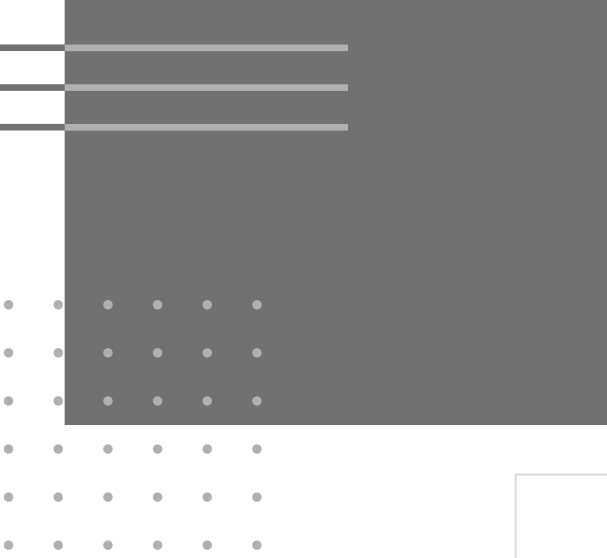
Develop an Energy-Aware Federated Learning Framework using Deep Reinforcement Learning (DQN) for intelligent client selection.

Specific Objectives

- Reduce battery consumption.
 - Minimize communication overhead.
 - Improve global model accuracy.
 - Reduce client dropouts.
 - Increase convergence speed.
 - Improve performance in low-connectivity regions.
- 

LITERATURE REVIEW

Title	Author(s) & Year	Description	Algorithm / Method Used	Metrics Evaluated	Demerits / Research Gap
Communication-Efficient Learning of Deep Networks from	Brendan McMahan et al. (2017)	Introduced Federated Learning where clients train locally and share	FedAvg	Accuracy, Communication Rounds	Random client selection, ignores battery, bandwidth, and device heterogeneity. (Frontiers)
Federated Optimization in Heterogeneous Networks (FedProx)	Tian Li et al. (2020)	Improved FL for heterogeneous devices by reducing client drift	FedProx	Accuracy, Loss, Convergence	Does not consider energy consumption or network reliability. (Frontiers)
Federated Learning with Non-IID Data	Yue Zhao et al. (2018)	Investigated the impact of non-IID data on federated learning	FedAvg + Data Sharing	Accuracy, Convergence	Requires additional data sharing; no energy optimization. (Frontiers)
Adaptive Federated Learning in Resource-Constrained Edge	Shiqiang Wang et al. (2019)	Adaptive scheduling for FL under edge resource constraints	Adaptive Federated Learning	Communication Cost, Latency, Accuracy	Limited client selection strategy and no reinforcement learning. (MDPI)
Client Selection for Federated Learning with Heterogeneous	Takayuki Nishio & Ryo Yonetani (2019)	Selected clients based on available computational resources	FedCS	Training Time, Accuracy	Static resource scheduling; battery and packet loss ignored. (Taylor & Francis)
Ed-Fed: Resource-Aware Client Selection for Edge Devices	Zitha Sasindran et al. (2023)	Proposed a framework with resource-aware client selection to reduce	Resource-Aware Client Selection	Waiting Time, Communication Delay	Focuses mainly on resource scheduling; no energy-aware reinforcement learning. (arXiv)
FedLE: Federated Learning Client Selection with Lifespan Extension	Jiajun Wu et al. (2023)	Energy-efficient client selection that extends battery life of IoT	FedLE (Battery-Aware Selection)	Battery Lifetime, Accuracy, Training Rounds	Does not consider bandwidth or packet loss together with battery. (arXiv)
Eco-FL: Enhancing Federated Learning Sustainability through	Martina Savoia et al. (2024)	Reduced energy consumption in edge FL through efficient client	Eco-FL	Energy Consumption, Accuracy, Communication Cost	Limited adaptation to changing network conditions. (ScienceDirect)
Energy Efficient Client Selection in Federated Learning for Orbital Edge	Bara'ah Al-Blewi et al. (2025)	Reinforcement learning-based client selection for energy-efficient FL in	FedSCS + Reinforcement Learning	Energy Consumption, Accuracy, Training Time	Designed specifically for satellite/LEO environments rather than terrestrial mobile edge networks. (Western Sydney University)
Deep Q-Network for Human-Level Control	Volodymyr Mnih et al. (2015)	Introduced Deep Q-Network (DQN), enabling reinforcement learning	Deep Q-Network (DQN)	Reward, Learning Efficiency	Not designed specifically for federated learning or edge computing. (Frontiers)

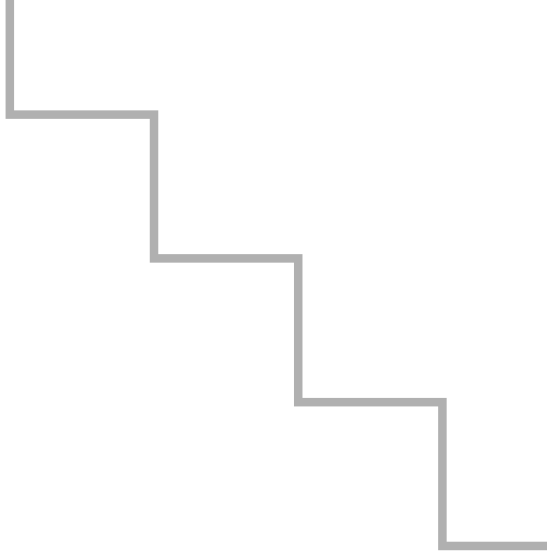


ANALYSIS

Existing System	Proposed System
Random client selection	Intelligent DQN-based client selection
High energy consumption	Energy-aware scheduling
High communication overhead	Reduced communication cost
Frequent client dropouts	Reliable client participation
Slow convergence	Faster convergence
Poor performance in unstable networks	Optimized for weak networks



PROPOSED METHODOLOGY



Data Preparation

- Select benchmark datasets (MNIST / CIFAR-10)
- Distribute data among edge devices using IID / Non-IID (Dirichlet Distribution)

Edge Device Registration

- Android edge devices register with the FastAPI Server
- Device telemetry is collected:
 - Battery Level
 - Bandwidth
 - Packet Loss
 - Device Reliability

Intelligent Client Selection

- Telemetry data is sent to the Deep Q-Network (DQN) Agent
- DQN evaluates device conditions
- Selects the most suitable clients for training

Local Model Training

- Selected clients train the CNN model locally
- User data remains on the device
- Only model parameters are generated

Federated Aggregation

- Local model weights are transmitted to the FastAPI Server
- FedAvg aggregates local updates
- Global model is updated

Global Model Distribution

- Updated global model is sent back to all participating clients
- Process repeats until desired accuracy is achieved

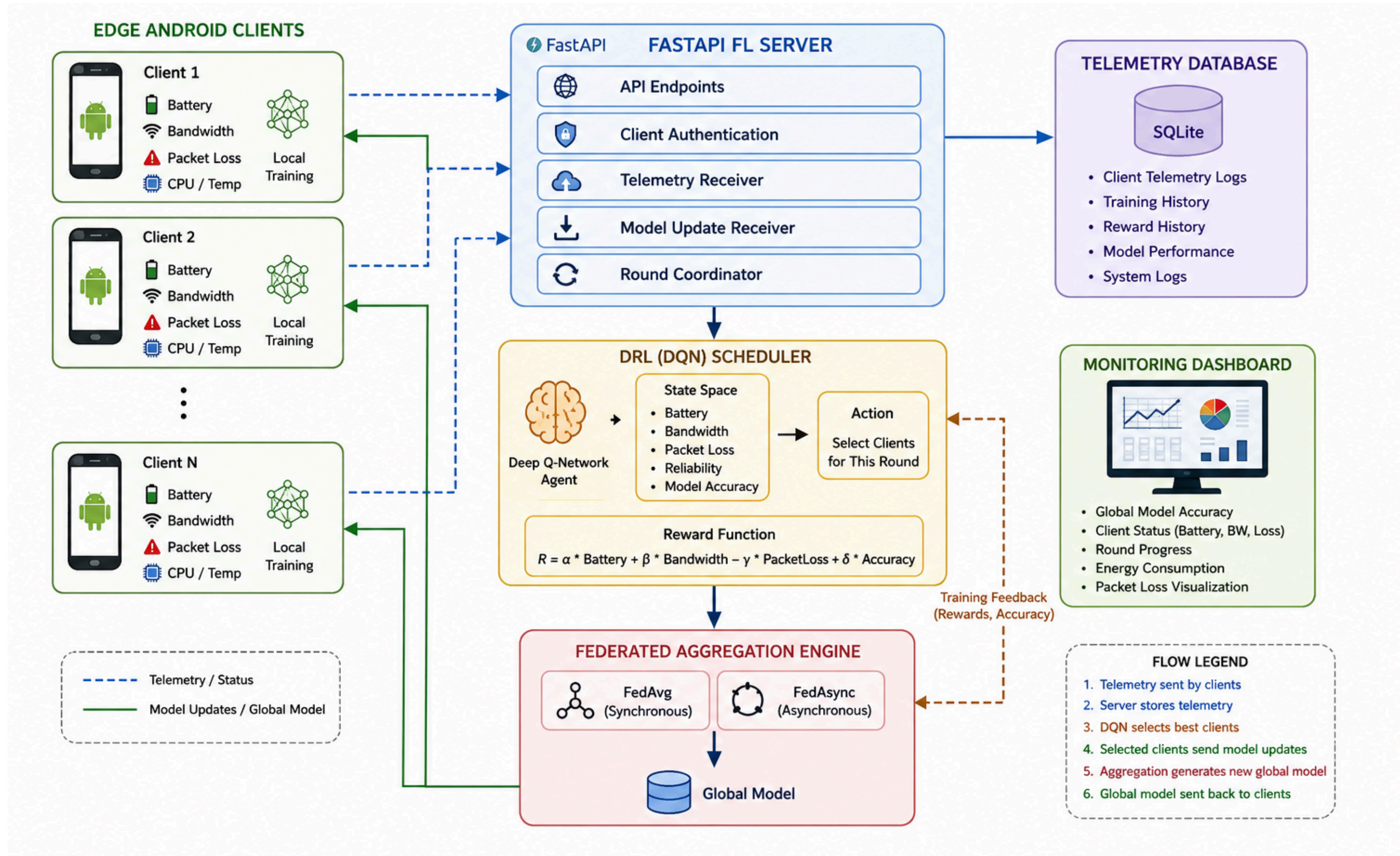
Performance Evaluation

Evaluate the proposed system using:

- Model Accuracy
- Energy Consumption
- Communication Overhead
- Convergence Speed
- Packet Loss Resilience

The proposed methodology integrates Deep Reinforcement Learning with Federated Learning to intelligently select energy-efficient edge devices, perform privacy-preserving local training, and improve global model performance in low-connectivity environments.

System Architecture



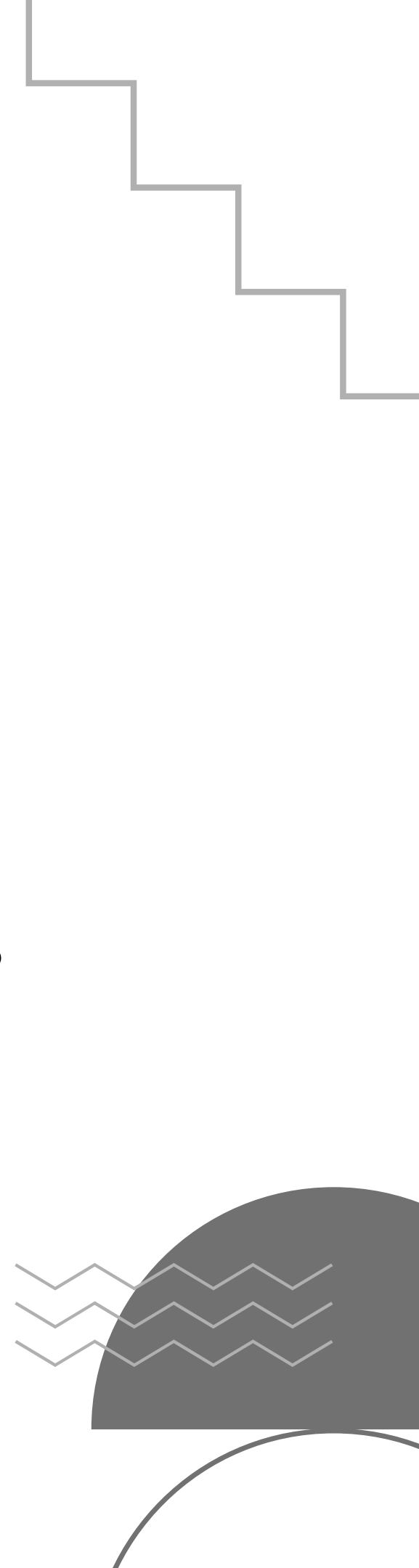


EXPECTED OUTPUT / PROJECT OUTCOME

Android Application

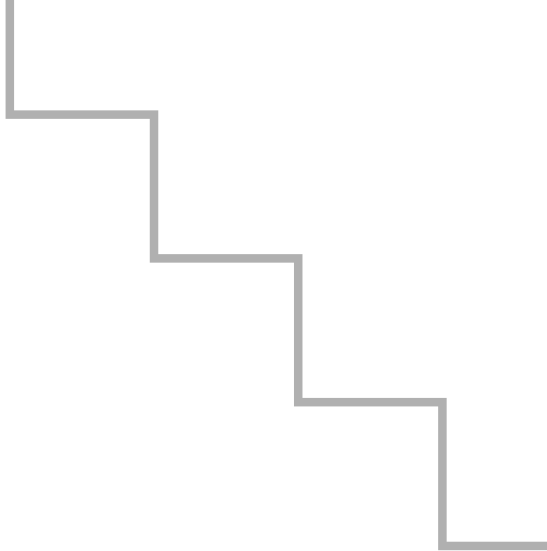
- Local model training
- Battery monitoring
- Network monitoring
- Automatic telemetry upload

Server Dashboard

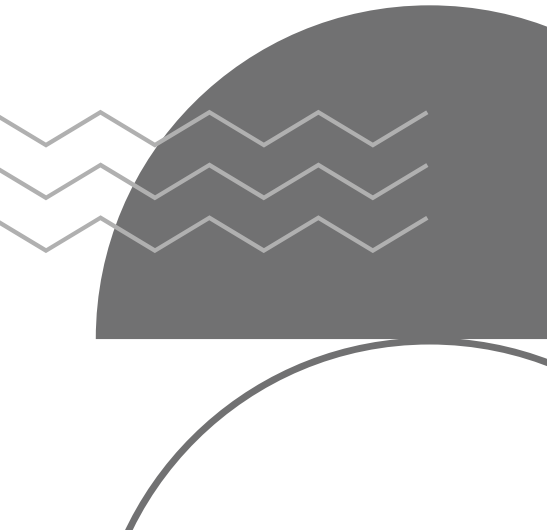
- Connected clients
 - Training rounds
 - Battery statistics
 - Accuracy graph
 - Packet loss monitoring
- 



Performance Improvements



Metric	Expected Improvement
Model Accuracy	95–97%
Battery Consumption	25–35% reduction
Communication Cost	20–30% reduction
Client Dropout	35–45% reduction
Convergence Speed	1.5–2× faster



TECHNOLOGIES USED

Software

- Python
- PyTorch
- FastAPI
- Android Studio
- Kotlin
- SQLite
- Chart.js

Hardware

- Android Smartphones
- Laptop/Desktop
- Wi-Fi / Mobile Network

Machine Learning

- Federated Learning
- Deep Reinforcement Learning
- Deep Q Network (DQN)

Future Scope

Future Enhancements

- Differential Privacy
- Secure Aggregation
- Blockchain-enabled Federated Learning
- Multi-Agent Reinforcement Learning
- TinyML Deployment
- Federated Transfer Learning
- 5G Network Slicing
- Real-time IoT Deployment

References

1. [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in Proc. 20th Int. Conf. Artificial Intelligence and Statistics (AISTATS), Fort Lauderdale, FL, USA, 2017, pp. 1273–1282.
2. [2] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated Optimization in Heterogeneous Networks," in Proc. Machine Learning and Systems (MLSys), Austin, TX, USA, 2020.
3. [3] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated Learning with Non-IID Data," arXiv preprint arXiv:1806.00582, 2018.
4. [4] T. Nishio and R. Yonetani, "Client Selection for Federated Learning with Heterogeneous Resources in Mobile Edge," in Proc. IEEE Int. Conf. Communications (ICC), Shanghai, China, 2019.
5. [5] S. Wang, T. Tuor, T. Salonidis, K. Leung, C. Makaya, T. He, and K. Chan, "Adaptive Federated Learning in Resource-Constrained Edge Computing Systems," IEEE Journal on Selected Areas in Communications, vol. 37, no. 6, pp. 1205–1221, Jun. 2019.
6. [6] V. Mnih et al., "Human-Level Control Through Deep Reinforcement Learning," Nature, vol. 518, no. 7540, pp. 529–533, Feb. 2015.
7. [7] Z. Sasindran, R. M. Venkatesan, and S. Chatterjee, "Ed-Fed: Resource-Aware Client Selection Framework for Federated Learning in Edge Computing," arXiv preprint arXiv:2307.07199, 2023.
8. [8] J. Wu, Y. Zhang, X. Liu, and H. Wang, "FedLE: Federated Learning Client Selection with Lifespan Extension for Edge IoT Networks," arXiv preprint arXiv:2302.07305, 2023.
9. [9] M. Savoia, F. Restuccia, and T. Melodia, "Eco-FL: Enhancing Federated Learning Sustainability Through Energy-Efficient Client Selection," Computer Networks, vol. 243, Art. no. 110315, 2024.
10. [10] B. Al-Blewi, A. Al-Fuqaha, and M. Guizani, "Energy-Efficient Client Selection in Federated Learning for Orbital Edge Computing," in Proc. IEEE Global Communications Conference (GLOBECOM), Cape Town, South Africa, 2025.

Thank You